Do the breaks the app teaches for each pawn structure hold at his level?

Each structure's breaks against his own move in 909 of his quiet positions, refereed by Stockfish, with strong players' positions beside them

Chess Coach research lab

Abstract

When the app names the pawn structure in one of his positions and a break its catalogue teaches for that structure can be played, is the break a sound move there, more often than the move he played? In 909 quiet middlegame and endgame positions from his 10-minute games, Stockfish at 250,000 nodes calls a move sound when it gives up no more than 3 win% against the best. By the rule fixed before any move was scored, the break misleads: the breaks are sound 20.5% and 22.4% of the time on the two halves of his games, his own moves 55.2% and 57.9%, a gain of −34.7 and −35.5 points with both intervals below 0, and both controls pass. A break that can be played is sound about as often as a random move that is not a break, and gives up 15.4 and 15.4 win% on average against his moves' 5.6 and 5.5. It is so in every structure with enough positions, and in 1,200 positions of players rated 2200 or more, whose breaks are sound 22.8% of the time against their own moves' 68.8%: a break is rarely right merely because it can be played. Added after the run: the breaks he played were sound 58.2% of the time, the ones he passed over 18.5%, and where a break is Stockfish's best, in 10.3% of these positions, he played one 25.5% of the time against 6.7% elsewhere. The app should not offer a structure's break as the plan whenever it is on; what it can teach is the moment, with the engine's word on each position.

1 The question

A coach explains a middlegame by its pawn structure. In the Carlsbad the plan is the minority attack and its break is b4-b5; against the London pyramid, Black strikes with . . . c5. The app does the same. It names the structure on the board, teaches each structure's plans and breaks from its catalogue, and its structure index counts the games in which he played the structure's break. This study asks whether that teaching holds in his games: when a break his structure teaches can be played in one of his quiet middlegame or endgame positions, is it a sound move there, more often than the move he played?

If the break holds, the review and the drills can teach the structure's break as the plan when it is on, and the count of breaks played measures something worth raising. If it misleads, playing the break whenever it can be played is worse than his own choice, and the app has to teach when a break is right, not only which break.

This study follows the one on positions like his. There, strong players' moves from positions like his carried over weakly, and best where the app names the pawn structure, which suggested testing the plan a structure calls for in place of the next move from a similar game.

2 How it was measured

His positions. His 647 10-minute games in the app's archive at commit `f4bd63f19b`, in order of their end time: the first 324 (July 22 to August 23) and the last 323 (August 23 to October 15) are the two halves. Nothing is fitted, so the halves are two replications. The positions are the ones the study on positions like his drew: before one of his moves, past the opening by the app's divider, not in check, with at least two legal moves and eight men or more, his win% before the move between 10 and 90 by the archive's evaluation, and his clock at least 30 seconds, 8,896 positions in all. The app names a structure in 2,411 of them (27%). A position enters this study when a break is on: at least one of his legal moves is a break of that structure for his side. That leaves 1,359 positions from 245 games, 730 in the first half and 629 in the second. In 1,040 positions with a named structure no break was on, and in twelve the catalogue gives his side no break.

The breaks. The app's catalogue writes each structure's breaks for each side as pawn moves, such as c3-c4 (Table 1). For three structures, the isolated queen's pawn, hanging pawns and the queenside majority, it writes them from the holder's side, the side whose pawns make the structure, and the app reads them with the ranks mirrored when Black holds it. A move is a break when the app's own reading counts it, played after the game's moves so far: a pawn move of his to the last square a break names, a capture landing there included. The structure is the one the app names on the board of the position.

Strong players' positions. The study on positions like his streamed the first 400,000 rated games of Lichess's September 2026 database (CC0) between two players rated 2200 or more. Read in file order, each game offers the positions before a move by either side that are past the opening, not in check, with eight men or more, at least two legal moves, the mover's clock at least 30 seconds and a structure the app names. A seeded hash of the game shuffles them, and the first of up to three with a break on for the mover is the game's one position. The first 11,392 games gave 3,000 positions; 5,219 games had no position with a named structure, and in 3,173 none of the tries had a break on. Most are blitz: 2,781 blitz, 209 rapid and 10 classical. They are scored in order until 1,200 pass the referee's quiet filter.

The moves compared. In each of his positions, **B** is every legal move of his that is a break, and B's soundness is the share of them that are sound. **A** is his move. **N** is Maia-2's most likely move at his band and **U** at the band above: his band is his rating at the game with the app's offset of 250 points, the band above the same of his rating plus 200, each conditioned on the opponent's band. **R** and **R2** are two legal moves that are not breaks, each drawn uniformly at random with a seed, independently of each other. In the strong players' positions: B, **S**, the strong player's move, R and R2.

The referee. Stockfish through the app's engine service at the grading budget of 250,000 nodes, with its hash cleared before each position and no tablebase, gives the best move and scores every move compared. A move is *sound* when it gives up no more than 3 win% against the best, by the app's own test for an alternative move. A position is *quiet* when the best move is not a capture, a check or a promotion and the win% it gives the side to move is between 10 and 90; only quiet positions are scored. A score that is not exact counts as missing, and a position with one leaves the analysis.

The measures. The *gain* is B's soundness minus A's, averaged over positions, paired by position; where his move was a break, B and A share it. 2,000 bootstrap draws with a fixed seed resample his games within each half, or the strong players' positions, and intervals are the 95% percentile intervals.

The rule, fixed before any move was scored. Two controls come first, and if either fails the study gives no verdict.

1. **The referee tells moves apart here.** His own move's soundness minus R's must have its interval above 0 on each half, and the strong player's move's minus R's on the strong players' positions.
2. **The null.** R2 against R, in place of B against A, must read neither as the break holding nor as it misleading.

Then, on his positions, *the break holds* when the gain's interval lies above 0 on both halves, *the break misleads* when it lies below 0 on both, and it is *no better than his own move* otherwise. A half with fewer than 150 scored positions is *not enough data*, whatever the numbers say. The rule and the scripts went to the lab before any move was scored; the positions with a break on had been drawn and counted, with no move scored.

3 Coverage

	The breaks the app teaches		His positions		Strong players'	
Structure	White	Black	Break on	Scored	Drawn	Scored
London pyramid	c3-c4, e3-e4	… c7-c5, … e6-e5	638	427	438	178
Isolated queen's pawn*	d4-d5	… b5-b4 to remove a defender	252	144	1,171	408
Reversed Carlsbad	f4-f5, c3-c4	… b5-b4, … e6-e5	208	153	215	100
Advance chain	f4-f5	… c7-c5, … f7-f6	195	136	400	171
Carlsbad	b4-b5, e3-e4	… c6-c5, … f7-f5	30	23	197	102
Pirc set-up	e4-e5, f4-f5, h4-h5	… c7-c5, … e7-e5	18	11	229	96
King's Indian chain	c4-c5	… f7-f5, … g5-g4	11	9	224	96
Hanging pawns*	d4-d5, c4-c5	–	7	6	126	49

Table 1: The app's structures with breaks, the breaks its catalogue teaches each side, and the positions with a break on: his, both halves, and the strong players' positions drawn, before the referee's quiet filter, and those scored. *Written from the holder's side: White's column is the holder's breaks and Black's the other side's, and the app reads both with the ranks mirrored when Black holds the structure.

A break is on in 56% of his positions with a named structure (Table 1). 47% of them are the London pyramid, 601 of its 638 with him as White, and nearly all of his reversed Carlsbad and advance chain positions have him as Black. The strong players' positions are most often the isolated queen's pawn, 39% of those drawn.

The referee keeps the quiet ones: 491 positions from 114 games in the first half and 418 from 103 games in the second. In 233 and 208 positions the best move is a capture, a check or a promotion, and in 6 and 3 the referee reads the game as decided. 1,824 strong players' positions were tried

before 1,200 were quiet. Every score was exact.

4 The controls

The referee tells moves apart. His own move is sound 36.3 and 37.6 points more often than a random move that is not a break (30.8 to 42.0; 31.7 to 43.4), and the strong player's move 51.7 points more often (48.6 to 54.7). All three intervals lie above 0.

The null passes. R2 against R reads −1.4 points (−5.6 to 2.9) in the first half and −2.4 points (−6.2 to 1.6) in the second; both intervals hold 0.

5 The verdict: the break misleads

	His, first half	His, second half	Strong players
Positions (games)	491 (114)	418 (103)	1,200 (1,200)
B, the breaks	20.5 (15.9, 25.2)	22.4 (17.2, 27.8)	22.8 (20.5, 25.1)
A, his move; S, the strong player's	55.2 (50.3, 60.1)	57.9 (52.0, 63.5)	68.8 (66.1, 71.4)
N, Maia-2 at his band	66.0 (61.6, 70.6)	71.5 (66.4, 76.7)	–
U, the band above	67.6 (62.9, 72.3)	73.0 (67.9, 78.2)	–
R, a random move	18.9 (14.6, 23.2)	20.3 (15.3, 25.6)	17.0 (14.9, 19.2)
R2, a second random move	17.5 (13.6, 21.4)	17.9 (13.2, 22.8)	17.6 (15.4, 19.7)
Gain: B minus A, or S (points)	−34.7 (−40.8, −28.5)	−35.5 (−42.1, −28.7)	−46.0 (−49.2, −42.8)
B minus N (points)	−45.5 (−50.7, −39.8)	−49.2 (−56.4, −42.3)	–
Control: A, or S, minus R (points)	36.3 (30.8, 42.0)	37.6 (31.7, 43.4)	51.7 (48.6, 54.7)
Null: R2 minus R (points)	−1.4 (−5.6, 2.9)	−2.4 (−6.2, 1.6)	–

Table 2: The share of each arm's moves that are sound, in %, and the differences the rule reads, in points, with 95% intervals resampling his games, or the strong players' positions. A move is sound when it gives up no more than 3 win% against Stockfish's best.

The breaks are sound 20.5% and 22.4% of the time on the two halves, his own moves 55.2% and 57.9% (Table 2). The gain is −34.7 points (−40.8 to −28.5) and −35.5 points (−42.1 to −28.7), below 0 on both halves, so by the rule the break misleads: playing the structure's break whenever it can be played would be worse than his own choice.

A break that can be played is sound about as often as a random move that is not a break, at 18.9% and 20.3%; paired by position, the breaks are 1.5 points (−3.5 to 6.6) and 2.0 points (−3.8 to 7.8) above it (Table 7). Being one of the structure's breaks adds nothing to a move's chance of being sound when all that is known is that it can be played. Maia-2's most likely move at his band is sound 66.0% and 71.5% of the time, and at the band above 67.6% and 73.0%, more often than his own.

	His, first half	His, second half	Strong players
Win% given up: the breaks	15.4 (13.5, 17.3)	15.4 (13.6, 17.2)	13.4 (12.7, 14.1)
Win% given up: the move played	5.6 (4.8, 6.5)	5.5 (4.5, 6.7)	3.1 (2.8, 3.4)
The breaks that are the best (%)	8.5 (5.5, 11.8)	6.0 (3.8, 8.5)	8.8 (7.4, 10.4)
He, or they, played a break (%)	9.2 (6.7, 12.0)	8.1 (5.8, 10.7)	11.7 (9.8, 13.5)
Maia-2 at his band gives the breaks (%)	7.5 (6.1, 9.2)	6.1 (4.9, 7.5)	–
At the band above (%)	7.1 (5.7, 8.9)	5.9 (4.7, 7.3)	–

Table 3: The mean win% the breaks and the move played give up against Stockfish's best (a move that scores above the best gives up none), the share of breaks that are the best, and how often a break is found: how often the move played was a break, and the probability Maia-2 gives the breaks together at his band and at the band above. 95% intervals.

6 What the positions describe

How much a break gives up. On average the breaks give up 15.4 and 15.4 win% against the best, about three times what his moves give up, 5.6 and 5.5 (Table 3). 8.5% and 6.0% of the breaks are the referee's best.

Finding it. He played a break in 9.2% and 8.1% of these positions, and Maia-2 at his band gives the breaks together 7.5% and 6.1% of its probability, at the band above 7.1% and 5.9%. Strong players played a break in 11.7% of theirs.

Strong players. In 1,200 positions of players rated 2200 or more, mostly blitz, the breaks are sound 22.8% of the time against the strong player's move's 68.8%, −46.0 points (−49.2 to −42.8), and a random move that is not a break 17.0%; paired by position, their breaks are 5.8 points above it (2.9 to 8.6). The break that can be played is rarely the move at their level either, and trails their own move by more than his trails his, so what misleads is the reading of a break as the plan whenever it is on, not his level.

7 By structure, phase and chance

Structure	Positions	Games	B (%)	A (%)	Gain (points)
London pyramid	427	111	27.6 (22.2, 33.3)	56.4 (51.0, 61.9)	−28.8 (−34.9, −22.4)
Reversed Carlsbad	153	38	8.2 (2.1, 16.5)	57.5 (48.9, 66.0)	−49.3 (−60.9, −37.9)
Isolated queen's pawn	144	48	22.9 (16.1, 29.7)	54.2 (46.0, 62.5)	−31.3 (−40.9, −21.2)
Advance chain	136	32	12.5 (6.2, 20.3)	57.4 (48.7, 66.4)	−44.9 (−54.5, −34.8)
Carlsbad	23	7	not enough data ($n = 23$)		
Pirc set-up	11	9	not enough data ($n = 11$)		
King's Indian chain	9	2	not enough data ($n = 9$)		
Hanging pawns	6	3	not enough data ($n = 6$)		

Table 4: His positions by structure, both halves together: the share of the breaks and of his moves that are sound, and the gain, with 95% intervals, for each structure with at least 30 scored positions.

Every structure. The London pyramid's breaks are sound 27.6% of the time against his move's 56.4%, the isolated queen's pawn's 22.9% against 54.2%, the reversed Carlsbad's 8.2% against 57.5%, and the advance chain's 12.5% against 57.4% (Table 4). In each structure with enough positions the gain's interval lies below 0: it is not only the London.

His positions	Positions	Games	B (%)	A (%)	Gain (points)
Middlegame	819	209	20.3 (17.1, 23.9)	56.3 (52.3, 60.3)	−36.0 (−40.4, −31.3)
Endgame	90	29	30.6 (16.2, 43.5)	57.8 (45.9, 68.3)	−27.2 (−41.3, −14.2)
The first chance in a game	172	172	22.1 (16.9, 27.9)	61.0 (53.5, 68.0)	−39.0 (−47.4, −30.5)
Later chances	737	175	21.2 (17.3, 25.0)	55.4 (51.3, 59.4)	−34.2 (−39.3, −29.2)
His move not a break, first half	446	108	17.2 (12.9, 21.9)	54.9 (49.9, 60.2)	−37.8 (−44.4, −31.1)
His move not a break, second half	384	96	20.1 (14.9, 25.5)	57.8 (52.1, 63.4)	−37.8 (−44.6, −31.0)

Table 5: His positions by the divider's phase and by whether the position is the first in its game with a break on, both halves together, and his positions where his move was not a break, by half: the share of the breaks and of his moves that are sound, and the gain, with 95% intervals.

Phase, the first chance, and the breaks he passed over. In the middlegame the breaks are sound 20.3% of the time against his move's 56.3%, and in the endgame, in only 90 positions, 30.6% against 57.8% (Table 5). The first position in a game with a break on is no better for the break than the later ones, 22.1% against 21.2%. Where his move was not a break, the breaks he passed over are sound 17.2% and 20.1% of the time on the two halves.

8 Added after the run

These descriptions were added once his halves had been scored and looked at; none was named before the run, and they decide nothing.

Positions	n	Played a break	Played the best	Maia-2 gives the breaks, his band	The band above
His, a break is the best	94	25.5 (17.1, 33.7)	23.4 (14.8, 32.0)	19.5 (15.5, 24.7)	20.1 (15.9, 25.5)
His, no break is the best	815	6.7 (5.2, 8.4)	22.5 (19.6, 25.4)	5.4 (4.6, 6.2)	5.0 (4.2, 5.8)
Strong, a break is the best	132	42.4 (34.1, 50.8)	40.2 (31.8, 48.5)	–	–
Strong, no break is the best	1,068	7.9 (6.3, 9.5)	36.0 (33.1, 39.1)	–	–

Table 6: Added after the run. His positions, both halves together, and the strong players', where a break is Stockfish's best and where none is: how often the move played was a break and was the best, and the probability Maia-2 gives the breaks together. In %, with 95% intervals.

The moment. A break is Stockfish's best in 94 of his positions, 10.3% of them. There he played a break 25.5% of the time (17.1 to 33.7), against 6.7% where no break is the best, and Maia-2 at his band gives the breaks 19.5% against 5.4% (Table 6). He and his band read when a break is due, and both miss it most of the time. Strong players played a break 42.4% of the time where one is the best, in 132 positions, and 7.9% elsewhere.

Positions	n	Some break sound	The move played, sound	Difference (points)	B minus R (points)
His, first half	491	26.3 (21.0, 31.9)	55.2 (50.3, 60.1)	−28.9 (−35.5, −21.9)	1.5 (−3.5, 6.6)
His, second half	418	28.0 (21.4, 34.9)	57.9 (52.1, 64.0)	−29.9 (−37.0, −22.4)	2.0 (−3.8, 7.8)
Strong players	1,200	26.3 (23.8, 28.9)	68.8 (66.1, 71.4)	−42.4 (−45.8, −39.0)	5.8 (2.9, 8.6)
		The break played, sound	The breaks, sound	The move played, sound	
His, played one	79	58.2 (47.4, 69.4)	51.3 (41.9, 61.3)	–	–
His, did not	830	–	18.5 (15.2, 22.0)	56.3 (52.3, 60.2)	–
Strong, played one	140	70.0 (62.1, 77.1)	60.0 (53.2, 66.8)	–	–
Strong, did not	1,060	–	17.8 (15.7, 20.0)	68.6 (65.8, 71.3)	–

Table 7: Added after the run. Above, the share of positions where at least one break is sound, against the share where the move played is, and the breaks' soundness minus a random move's, paired by position. Below, his positions, both halves together, and the strong players', where the move played was a break or was not: the share of the breaks played that are sound, of all the breaks, and of the moves played. In %, with 95% intervals.

His own choice. Where he played a break, in 79 positions, it was sound 58.2% of the time (47.4 to 69.4), about as often as his moves are; where he did not, the breaks he passed over were sound 18.5% (15.2 to 22.0). He breaks more selectively than the catalogue's reading of when a break is on (Table 7). Strong players do the same: the breaks they played, in 140 positions, were sound 70.0% of the time, the ones they passed over 17.8%. At least one break is sound in 26.3% and 28.0% of his positions, against 55.2% and 57.9% for his move.

Each break apart. His most common breaks are the London pyramid's c3-c4, sound 29.0% of the time in 373 positions against his move's 56.6%, the London pyramid's e3-e4 (23.4% against 56.3%), the reversed Carlsbad's . . . e6-e5 (7.9% against 58.3%) and the advance chain's . . . f7-f6 (9.3% against 58.5%) (Table 8). No break with enough positions, his or the strong players', is sound as often as the move played. The strong players' most common is the isolated queen's pawn's d5-d4, sound 22.6% of the time against their move's 64.8%.

		His positions			Strong players'		
Structure	Break	n	The break	His move	n	The break	Their move
London pyramid	c3-c4	373	29.0 (22.6, 35.4)	56.6	63	25.4 (15.9, 36.5)	74.6
London pyramid	e3-e4	355	23.4 (17.7, 29.5)	56.3	80	17.5 (10.0, 26.3)	75.0
Reversed Carlsbad	. . . e6-e5	151	7.9 (2.0, 15.6)	58.3	48	12.5 (4.2, 22.9)	66.7
Advance chain	. . . f7-f6	118	9.3 (3.3, 18.1)	58.5	132	10.6 (6.1, 15.9)	62.1
Isolated queen's pawn	d5-d4	76	25.0 (16.0, 35.1)	50.0	159	22.6 (16.4, 28.9)	64.8
Isolated queen's pawn	d4-d5	44	11.4 (2.3, 21.1)	50.0	143	22.4 (16.1, 29.4)	68.5
Carlsbad	e3-e4	22	–	–	55	12.7 (5.5, 21.8)	70.9
London pyramid	. . . c7-c5	22	–	–	39	28.2 (15.4, 43.6)	71.8
Advance chain	. . . c7-c5	18	–	–	37	32.4 (18.9, 48.6)	70.3
Isolated queen's pawn	. . . b4-b5 to remove a defender	16	–	–	42	31.0 (16.7, 45.2)	73.8
London pyramid	. . . e6-e5	12	–	–	66	15.2 (7.6, 24.2)	72.7
King's Indian chain	. . . f7-f5	9	–	–	51	43.1 (29.4, 56.9)	70.6
Isolated queen's pawn	. . . b5-b4 to remove a defender	8	–	–	64	25.0 (15.6, 35.9)	60.9
Pirc set-up	. . . e7-e5	8	–	–	32	34.4 (18.8, 50.0)	75.0
Pirc set-up	e4-e5	3	–	–	47	23.4 (12.8, 36.2)	78.7
Reversed Carlsbad	c3-c4	2	–	–	50	14.0 (6.0, 24.0)	74.0
Pirc set-up	. . . c7-c5	2	–	–	36	38.9 (25.0, 55.6)	77.8
Carlsbad	. . . c6-c5	1	–	–	47	21.3 (10.6, 34.0)	76.6
King's Indian chain	c4-c5	0	–	–	36	33.3 (19.4, 50.0)	55.6

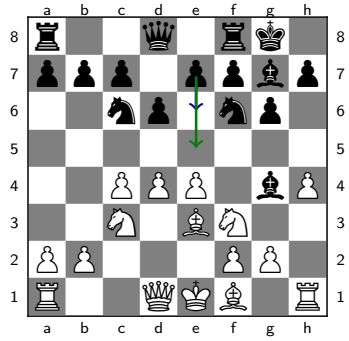
Table 8: Added after the run. Each break the catalogue teaches, apart, where it is on in at least 30 of his positions (both halves) or of the strong players': the positions, the share of the break's moves that are sound, with its 95% interval, and the share of the moves played there that are, in %. Each break is the catalogue's text as the app reads it for the side to move: for a structure written from the holder's side, with the ranks mirrored when Black holds the structure.

Structure	Positions	B (%)	S (%)	B minus S (points)
Isolated queen's pawn	408	23.8 (19.9, 27.9)	66.4 (62.0, 70.8)	−42.6 (−48.0, −37.0)
London pyramid	178	19.7 (14.9, 25.0)	72.5 (65.7, 79.2)	−52.8 (−60.1, −44.7)
Advance chain	171	15.2 (10.5, 20.5)	65.5 (58.5, 72.5)	−50.3 (−58.8, −42.1)
Carlsbad	102	16.7 (9.8, 23.5)	73.5 (64.7, 82.4)	−56.9 (−66.7, −46.6)
Reversed Carlsbad	100	15.0 (9.0, 22.0)	71.0 (62.0, 80.0)	−56.0 (−68.0, −43.5)
King's Indian chain	96	38.0 (28.1, 47.9)	63.5 (54.2, 72.9)	−25.5 (−37.5, −13.5)
Pirc set-up	96	33.9 (25.0, 42.7)	77.1 (68.8, 85.4)	−43.2 (−55.2, −31.3)
Hanging pawns	49	28.6 (21.4, 35.7)	65.3 (51.0, 79.6)	−36.7 (−51.0, −22.4)

Table 9: Added after the run. The strong players' positions by structure: the share of the breaks and of their moves that are sound, and the difference, with 95% intervals, for each structure with at least 30 scored positions.

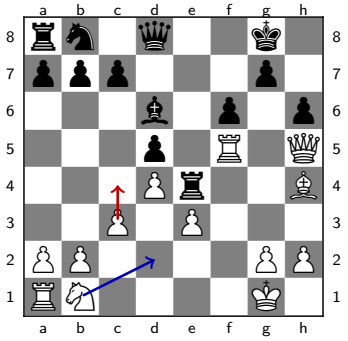
9 Two positions

Two of his positions show what a break that can be played looks like (Figure 1): the first of the second half where a break is the best move and he played neither a break nor another sound move, and the first London pyramid position of the second half where every break is unsound and his own move is sound. They illustrate; they test nothing.



FEN `r2q1rk1/ppp1ppbp/2np1np1/8/2PPP1bP/2N1BN2/PP3PP1/R2QKB1R b KQ - 2 8`

(a) He is Black, in the middlegame, in the Pirc set-up. The best move is 8. . . e5, at 55% to win. The break: 8. . . e5 (sound, 55%). He played 8. . . e6 (blue), not sound at 49%.



FEN `rn1q2k1/ppp3p1/3b1p1p/3p1R1Q/3Pr2B/2P1P3/PP4PP/RN4K1 w - - 0 16`

(b) He is White, in the middlegame, in the London pyramid. The best move is 16. ♘d2, at 61% to win. The break: 16. c4 (not sound, 54%). He played 16. ♘d2 (blue), sound at 61%.

Figure 1: Two of his positions with a break on. A break is green where it is sound and red where it is not; win% is his after each move. Stockfish at 250,000 nodes, rerun for this figure.

10 What the app can take

- **Not the break as the plan whenever it is on.** In his positions a break that can be played is sound about one time in five, no more often than a random move, while his own move is sound more than half the time. When the app names a structure's break in a position, in the review, a drill or the coach's reading of the position, it says whether Stockfish finds the break sound there, and does not offer it as the plan otherwise.
- **The count of breaks played is no target.** The structure index counts the games where he played the break. A break played merely because it can be is mostly unsound, so a higher count is not progress by itself; what the index can count instead is the break played where it was sound, and the break missed where it was the best.
- **Teach the moment.** Where a break is the best move, about one position in ten here, he played one about one time in four. Those positions, and the ones where a break was on and wrong, are the material for a drill on when to break, each checked by Stockfish and each with the structure's trap, which the catalogue already writes. A next round can test whether the conditions the app's coaching method asks before a break (pieces pointed at the lines it opens, rooks ready for the file, enough force on that side) tell when a break is sound, with its rule fixed first.

11 Caveats

- **A break stands for a plan.** The break is tested as the next move. The preparation a plan asks for first, rooks behind the advance or the trap each structure names, is not tested, except as the first chance in a game against the later ones.
- **Breaks by square.** The app counts any pawn move to a break's last square as the break, a capture landing there included; this study keeps the app's reading.
- **The structure of the position.** The structure index names the structure a game settles into and holds; this study names it on the board of each position, as the app's reading of a position does.
- **Sound is a window.** In a quiet position several moves are often within 3 win% of the best, so B and A can both be sound; they are paired by position for that reason.
- **Many positions a game for him.** Every one of his positions that qualifies enters, so a game where a break stays on for many moves counts many times; intervals resample whole games for that reason. Strong players give one position a game.
- **Mostly the London, and mostly blitz.** About half of his positions are the London pyramid; the by-structure table shows the answer is not only the London's. Most strong games are blitz; his are rapid.
- **Added after the run.** The descriptions of the moment, his own choice, each break apart and the strong players' structures were chosen once his halves had been scored. They describe; a rule built on them needs positions this study did not see.